



Rialtas na hÉireann
Government of Ireland

Private Wires Policy Statement

July 2025

Prepared by the Department of
Climate, Energy and the Environment
gov.ie/DCEE

Table of Contents

Table of Contents.....	i
1 Introduction	2
2 Current Legal and Regulatory Position.....	2
3 Progress to Date	4
3.1 Public Consultation	4
3.2 Private Wires Guiding Principles.....	5
4 New Flexibilities for Businesses and Households.....	6
4.1 Private Lines from a Generator to a Customer.....	7
4.2 Hybrid Connections	8
4.3 On Street EV Charging	9
4.4 Expansion of Self Supply	9
5 Supporting Changes Needed	9
5.1 Standard Setting for Cables.....	10
5.2 Extending the Dial Before you Dig System	10
5.3 Charging Methodology.....	10
5.4 Expanding the Role and Resources of CRU	11
5.5 Internal Networks and Collector Cables	11
6 Next Steps	11

1 Introduction

The current legal and regulatory framework for electricity in Ireland provides that ESB Networks design, build own, operate and maintain the electricity distribution network, and as transmission system owner, they design build and maintain the onshore electricity transmission network. Many commentators and stakeholders have suggested that allowing private developers to construct, own and operate at least some onshore electricity lines between generators and customers would accelerate investment in electricity infrastructure, particularly investment in renewable electricity generation.

After extensive consultation the Government has decided to allow private wires to be built and operated in a limited number of circumstances where the public benefits of allowing such developments are clear. This decision can be revisited in future and the situations where private wires are permitted can be expanded should circumstances require in the future.

Our next step will be to bring legislation and supporting regulations to the Oireachtas to allow the construction of private wires in the following circumstances:

- Connections between a single generator and a separate single user of electricity, and/or between a battery and a separate single user of electricity where this is more cost effective for the customers in question and the system as a whole, than both customers connecting to the Grid;
- Where two separate legal entities are sharing a Grid connection, for example a solar farm and an electricity storage facility owned by different legal entities;
- To allow on street charging of Electric Vehicles; and,
- To allow a customer that self-supplies electricity to provide electricity to a separate customer in a contiguous premises.

2 Current Legal and Regulatory Position

The current regulatory framework for electricity infrastructure is based on the Electricity Regulation Act 1999 as amended. This Act defines four key roles in the Irish electricity system: transmission system operator (TSO); transmission system owner (usually referred to as the transmission asset owner or TAO); distribution system operator (DSO); and, distribution system owner (usually known as the distribution asset owner or DAO). The Act empowers the Commission for Regulation of Utilities (CRU) to licence undertakings to fill these roles. Accordingly, CRU has licensed EirGrid as the TSO, ESB as the TAO and DAO

and ESB Networks DAC as the DSO. As required under its TAO and DAO licences, ESB has designated a ring-fenced business unit known as ESB Networks to carry on the TAO and DAO functions. With the approval of the CRU these three network functions are operated together and are referenced collectively in this paper as “ESB Networks”.

The Act also defines “distribution” and “transmission”. Distribution is defined as:

“...the transport of electricity by means of a distribution system, that is to say, a system which consists of electric lines, electric plant, transformers and switchgear and which is used for conveying electricity to final customers”.

Transmission is defined as:

“...the transport of electricity by means of a transmission system, in the State or offshore, or both, that is to say, a system which consists, wholly or mainly, of high voltage lines and electric plant and which is used for conveying electricity from a generating station to a substation, from one generating station to another, from one substation to another or to or from any interconnector or to final customers but shall not include any such lines which the Board may, from time to time, with the approval of the Commission, specify as being part of the distribution system but shall include any interconnector owned by the Board”.

The effect of this legal framework is that only ESB Networks can own a line that brings electricity from an onshore generator to an electricity customer. In addition, only ESB Networks DAC can operate the lines that make up the distribution system. EirGrid has the sole role in the operation of the lines that make up the transmission system and a legal ownership of offshore transmission assets.

There are two minor exceptions to this general rule against private ownership of electricity lines:

- A line connecting a generator or a customer to the Grid can be built on a “contestable” basis. In this case the generator, customer or contractors employed by them build the line which is then transferred to the ownership of ESB Networks when the line is energised.
- CRU can permit construction of a “direct line” but only in a situation where a customer has been refused a grid connection or where there is a dispute over the terms for a grid connection which has been referred to CRU for resolution.

3 Progress to Date

As noted above, some stakeholders have suggested that allowing customers to build their own “private wires” to connect electricity generation with demand could accelerate necessary investment in the Grid. These private wires are particularly associated with accelerating the roll out of renewable generation. The Climate Action Plan 2024 included a commitment for the Department of the Environment, Climate and Communications (DECC) to develop a Private Wires Policy Framework.

3.1 Public Consultation

The Department of the Environment, Climate and Communications published a consultation paper on Private Wires in August 2023. The consultation remained open until October 2023. A total of 128 responses were received from a wide range of individuals and groups including:

- Industry associations such as Wind Energy Ireland and the Irish Solar Energy Association;
- The system operators ESB Networks and EirGrid;
- Private energy developers; and,
- Individual citizens.

In July 2024 the Department of the Environment, Climate and Communications published a summary report of the results of the consultation exercise.

The role of Private Wires was seen by most respondents as a possible solution to issues experienced by electricity generators and customers in Ireland.

Respondents advised that Private Wires could be a solution to the reported slow grid connection process, benefiting customers, and allowing developers to take the scope for electrical infrastructure design and install into their own remit, and giving them project programme certainty. The key role identified among respondents was accelerating the rollout of renewable energy projects to help Ireland achieve its renewable electricity (RES-E) targets, however it was noted that if not appropriately regulated such a policy could see an increase in fossil fuel generation. Respondents agreed with the potential deployment opportunities identified as part of the consultation.

In terms of barriers, most respondents referred to the extensive range of regulation and standards that would need to be updated to permit Private Wires in Ireland, depending upon

the deployment opportunities permitted. The diverse range of potential scenarios of Private Wires demonstrates the extensive scope of the relevant areas to be examined when developing the policy framework and regulation of Private Wires. Multiple respondents noted that once the scope for Private Wires is more refined, further stakeholder engagement should take place.

Security of supply was also discussed as a risk, with the primary concern being that should Private Wire users have a national electricity grid connection, they would be drawing from the grid during times of low on-site generation, which is likely to coincide with peak demand on the grid. Others noted that the introduction of new generation assets by Private Wires to meet existing demand, could assist the overall electricity system. The introduction of Private Wires would also likely require a review of the recovery of network charges to ensure a fair balance of charges to customers. The risks of land-sterilising and infrastructure duplication were also flagged in relation to ensuring efficient development of the national electricity grid.

ESB Networks submitted that continued high levels of investment in the national electricity grid will be the mainstay of achieving Ireland's climate goals. They noted that significant investments and developments in the network are underway for the benefit of all customers, which may reduce the necessity and demand for private wires. ESB Networks noted their support of a policy around the use of Private Wires and consider that there may be limited circumstances in which private wires could play a useful role. Highlighting that, given the potential for Private Wires to impact on the national distribution and transmission systems, it is essential that a private wires policy is based on sound principles, does not give rise to unintended consequences, and that a proper regulatory impact assessment is conducted.

EirGrid welcomed the consultation and acknowledged the potential role Private Wires could play in accelerating the transition to Net Zero. Noting the importance, when developing a policy, of the potential benefits of Private Wires being quantified and balanced against related challenges and risks. The approach to Private Wires should be complementary to plan-led grid development, notably EirGrid's 'Shaping Out Electricity Future'. EirGrid noted that having a single Transmission System Operator (TSO) offers significant benefits in terms of safety, security, and reliability. While they are open to the potential for multiple asset owners, careful consideration must be given to how this would work in practice.

3.2 Private Wires Guiding Principles

In parallel with the publication of the consultation results in July 2024, Government approved a set of guiding principles for the development of the final Private Wires Policy. The statement on guiding principles set out that the new Private Wires Policy will aim to unlock

private sector resources to build new electricity infrastructure by expanding the right of private undertakings to connect supply directly with demand.

The statement set out that any expansion of the right of private developers to build Private Wires will be guided by the following policy considerations:

- The National Electricity Grid is a crucial piece of national infrastructure and will remain in public ownership.
- The objective of any change is to accelerate the deployment of renewable generation and storage.
- Any change must serve other policy objectives including security of supply, the offshore wind strategy, the hydrogen strategy, industrial policy, and the national climate objective.
- The National Electricity Grid will remain the preferred way to connect generators and customers of electricity. Where Grid-based solutions are available these will be preferred to Private Wires.
- Private Wires cannot undermine the financing of the National Electricity Grid. Those entities with a Private Wire must pay the full cost of the service they have sought from, or that is provided to them, by the National Electricity Grid.
- Private Wires will be built to the same technical and safety standards as the National Electricity Grid and will be capable of being taken in charge by the System Operators.
- Private Wires will not be permitted to undermine the efficient development of the National Electricity Grid e.g., by way of land sterilization.
- Private Wires will have to be developed with the consent of local communities and landowners. Private Wire developers will not have wayleave or other powers to use land.

4 New Flexibilities for Businesses and Households

Electricity generation, supply and demand in Ireland is organised as a network. All sources and users of electricity are connected to a common electricity Grid. If additional generating capacity is needed, or if thermal generation is being replaced by renewables, new generating units are connected to the Grid. These new generating units can then provide

electricity to potentially all users. Similarly, new customers demanding electricity connect to the Grid, giving them access to all providers of electricity.

This policy allows for the construction and operation of Private Wires in four specific scenarios where it is clear that allowing private investment in an electricity line is the most efficient approach to a new connection and is in the public interest. This change will require primary legislation, making detailed regulations and assigning new powers and tasks to the CRU, and potentially to other bodies. In particular safeguards will be needed to ensure that private wires are built and operated to the same standards of safety as the publicly owned grid. The end result of this will be a new opportunity for undertakings other than EirGrid and ESB Networks to build, own and operate electricity lines in the following four scenarios.

4.1 Private Lines from a Generator to a Customer

Consultation with stakeholders has indicated strongly that there are situations where it can be advantageous to the undertakings concerned, and the system as a whole, for a new source of generation to connect directly to a customer or another electricity undertaking. This private connection can also include battery storage. The nature of the supply and demand in question and the specific locations involved can make it more cost effective for two customers to connect directly to each other rather than both connecting to the main Grid, taking into account the costs for the customers and the public grid.

In future, if a customer wants to build and use a private line from a new source of electricity to an electricity customer and this can be done in line with the guiding principles already adopted, this will be allowed by legislation under certain circumstances. Ensuring that these new private lines are in the interest of the system as a whole and comply with the guiding principles will require a system of permitting. Customers wanting to build these private wires will have to seek permission from a CRU based on regulations which will ensure that:

- The project that is being facilitated serves public policy objectives such as security of supply, the offshore wind strategy, the hydrogen strategy, industrial policy, and the national climate objective;
- A lower cost, grid based, solution is not available to the customer or customers in question within a reasonable period of time;
- The System Operators will be able to recover the full cost of the services that they have contracted to provide to the electricity customer in question, including by means of revised tariffs if necessary;

- The Private Wire will be built to the same technical and safety standards as the National Electricity Grid and will be capable of being taken in charge by the System Operators;
- The Private Wires will not interfere with the operation of the Grid, or with any planned expansion or reinforcement of the Grid.

The developers of Private Wires will not enjoy any statutory powers to acquire or use lands. Private Wire projects will have to be built based on the agreement of any landowners affected.

4.2 Hybrid Connections

Hybrid Connections refers to a situation where more than one generation technology or electricity storage technology share a connection to the Grid. For example, a wind farm and an array of solar PV panels or a wind farm and a battery storage installation could share a single grid connection. Climate Action Plan 2023 included an action for CRU to “Ensure that hybrid technology grid connections are facilitated, and remaining barriers removed”.

Hybrid connections are efficient as they maximise the use of scarce grid infrastructure. They also accelerate the adoption of renewable energy and supporting technologies such as battery storage. The CRU’s work to carry out this action covers three areas:

- Removal of the Installed Capacity Cap (ICC) of 120% of Maximum Export Capacity (MEC);
- Facilitating the sharing of MEC behind the single connection point; and,
- Enabling Multiple Legal Entities (MLE) to share a single connection point.

CRU has already published a decision on the Installed Capacity Cap¹. It carried out a public consultation on Sharing the Maximum Export Capacity in March and April of this year. CRU is exploring the changes that may be needed to regulation and connection agreements to allow multiple legal entities to share a Grid connection. One issue that has emerged is that where two legal entities share a Grid connection this creates an electric line between them, and this may be in breach of the current legislation on Private Wires. This will be clarified and the legislation to implement this policy will remove this potential barrier to Hybrid Connections.

¹ CRU 202402 17th January 2024

4.3 On Street EV Charging

The Climate Action Plan target is for 30% of all private cars to be electric vehicles by 2030. Zero Emission Vehicles Ireland is a dedicated Office in the Department of Transport which, amongst other things, is working to ensure that the necessary public and private vehicle charging infrastructure is made available. A challenge has arisen in this work in the area of providing low-cost charging solutions for those without access to off-street parking, which could potentially include the running of electricity lines in public or shared spaces to enable charging. The legislation will be updated to provide clarity required for the provision of charging infrastructure for electric vehicles.

4.4 Expansion of Self Supply

The current regulatory structure for electricity allows for on-site generation. A customer is allowed to generate electricity for its own use and consume it on a single premises. No distribution or transmission of electricity takes place in this situation. Subject to technical and safety safeguards it will be useful to allow a customer to generate electricity and provide it to a separate customer on a contiguous site. The customers in question will remain subject to any connection agreements they have with ESB Networks.

5 Supporting Changes Needed

There are a number of complex regulatory and administrative issues that will have to be addressed in order to implement this policy. These will establish rules and processes to allow the four types of private wire described here to be built and operated in a safe and effective way. These issues are:

- Standard setting for cables;
- Regulation of competent personnel;
- Extending the dial before you dig system;
- Charging methodology;
- Expanding the role and resources of CRU;
- Internal networks/collector cables;
- Existing cables not in ESB Networks ownership; and,
- Clear rules and standards for how private wires will be incorporated into the operation and maintenance of the network, e.g. storm repairs or granting outages.

5.1 Standard Setting for Electricity Circuits

The guiding principles for Private Wires adopted by Government in July 2024 made it clear that any Private Wire will be built to the same technical and safety standards as the National Electricity Grid and will be capable of being taken in charge by the System Operators. All transmission and distribution lines are currently built to the standards set by ESB Networks for their own work and maintained according to appropriate maintenance cycles. These standards will have to be codified and published, and CRU will be given the powers and resources needed to ensure that Private Wires meet these standards. This will include ensuring that contractors undertaking validation work are properly qualified.

5.2 Extending the Dial Before you Dig System

ESB Networks operates a 'dial before you dig' service which allows contractors to contact it and other utility owners to check the location of underground services before breaking ground. However, only ESB Networks electricity cables are captured under this programme, and any cables that exist outside of the national grid, such as collector cables within an internal network for renewable generators, are not captured.

Once Private Wires are permitted a broader central mapping and recording of all electricity cables will be required to ensure the public are not at risk of coming across live electricity lines unexpectedly.

5.3 Charging Methodology

The System Operators, EirGrid and ESB Networks, recover their costs by charging electricity generators and customers (via their electricity suppliers) for their connection to the Grid. For larger users, these charges are a mix of a fixed annual amount based on the Maximum Import Capacity of the user's grid connection and a per unit amount based on the number of kilowatt/hours of electricity consumed. Where a new connection is built for a user ESB Networks will usually recover half of the cost from the user on connection and recover the rest through ongoing network charges.

Allowing private wires creates risks that the System Operators will not be able to recover their costs and that other customers may find themselves providing an effective cross subsidy to the private wire operator. If an electricity customer chooses to take all of their electricity via a private wire and ends their connection to the Grid there will be a risk that not all of the costs of providing the original Grid connection will be recovered by the ESB Networks. If an electricity customer using a Private Wire chooses to keep their Grid connection, the charges they will pay to the System Operators will be reduced by the

reduction in the amount of electricity that flows through the connection. However, the System Operators will be required to ensure that the connection can, at any time, provide the full Maximum Import Capacity of electricity flow to the customer, and will not see their costs reduce. This means that there will be reduced network costs for the private wire operator which will be recovered from other customers and moreover, the reserved network capacity cannot be allocated to future customers meaning that extra network reinforcement will be required to meet that demand, increasing cost for network customers

If these effects on the ability of System Operators are significant it will be necessary to reform the basis on which network connection charges are made, with an increased emphasis on one off charges to recover the fixed costs of connections and on charges based on the capacity of connections rather than the flow through a connection.

5.4 Expanding the Role and Resources of CRU

Allowing for private wires will create significant additional responsibilities for CRU, which will be responsible for permitting Private Wires, and ensuring that they are built and operated to the appropriate standards. The legal powers and resources of CRU will have to be increased appropriately. CRU will be required to engage with the system operators on the private wire applications as it is the SOs that will have the ability to consider how a private wire would impact on the development of the distribution or transmission system, how it would impact on delivery of government targets and provide assessment of any extra costs to the network customer.

5.5 Internal Networks and Collector Cables

The consultation process on Private Wires has highlighted that wind and solar farms often have extensive networks of cables to collect generated electricity from individual wind turbines or panels and bring them to a central point where the farm connects to the Grid. These wires can be lengthy, and in some cases cross roads and third party owned land. However, there are no publicly adopted standards or mechanisms to inspect such networks. The standards for design and safe operation for Private Wires will be extended to these internal networks as needed.

6 Next Steps

The next steps to implement this high-level statement of policy are:

- Primary legislation to amend the Electricity Regulation Act allowing for Private Wires will be brought to the Oireachtas:

- Supporting regulations to define standards and processes for granting permission for Private Wires will be adopted; and,
- The Commission for Regulation of Utilities will be resources to take on this new area of regulation.
- Continue to monitor industry developments e.g. the emerging concept of green energy parks to identify any future need for further liberalisation.